

Grand Performance Task: Smart Healthcare & Emergency Response Platform (SHERP)

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Course: BSIT – 3 / SEC – 1

1. Problem Scenario & Objectives

- **The Problem:** Current emergency response is too slow because medical data is trapped in silos. Patients often arrive at hospitals before their data does, costing lives.
 - **The Solution:** SHERP is a "Smart Platform" that uses wearable sensors and 5G to transmit vitals in real-time to ambulances and hospitals.
 - **Goals:** Reduce mortality through zero-latency data transmission and AR-assisted first response.
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2. System Architecture & Platform Integration

[DIAGRAM INSTRUCTION]: Draw four boxes stacked on top of each other:

1. **Top (Interaction):** AR Glasses & Doctor Dashboards.
 2. **Middle-Top (Convergence):** IoT Cloud, Blockchain, & Database.
 3. **Middle-Bottom (Networking):** 5G, Satellite, and M2M (Machine-to-Machine).
 4. **Bottom (Computing):** Wearable Sensors & Embedded RTOS.
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3. Platform Design Explanation

A. Computing Platform

- **Theory:** Uses **Distributed Computing** to process data across the wearable and the cloud.
- **OS:** Runs an **Embedded Real-Time OS (RTOS)** on the patient's device to ensure vitals are never missed.

- **Fault-Tolerance:** Uses **Checkpointing**; if the device reboots, it resumes tracking instantly.

B. Networking Platform

- **Infrastructure:** Uses **5G Broadband** for speed and **Satellite** for rural coverage.
- **M2M/Vehicular:** The patient's device talks directly to the ambulance (**V2X communication**) as it approaches.
- **Management:** Uses **SNMP protocols** to monitor network health and prevent outages.

C. Human & Media Platform

- **UI/UX:** A clean, high-contrast dashboard for surgeons.
- **AR Integration:** Paramedics use **AR Glasses** to see a digital overlay of the patient's medical history while treating them.
- **Visualization:** Uses **Big Data heatmaps** to show city-wide health trends.

D. Convergence Platform

- **IoT:** Integrates heart monitors and smart ambulances into one ecosystem.
- **Semantic Web:** Uses **RDF tagging** so different hospitals can share data even if they use different software.
- **Innovation:** A "Service-Oriented" model that connects health data directly to insurance for instant billing.

4. Security & Privacy Plan

- **Risk:** Man-in-the-Middle (MitM) attacks where hackers try to change medical data during transmission.
- **Protection:** 1. **Encryption:** AES-256 (the gold standard).

2. **Trust System:** **Blockchain** logs every time someone looks at a patient's file.

3. **Secure OS:** A **Microkernel** design that prevents a virus from taking over the whole device.

5. Implementation Considerations

- **Scalability:** Uses **Kubernetes** to handle millions of patients at once.

- **Real-time: Apache Spark** processes data streams instantly to detect heart attacks.
 - **Proof (Optional Logic):** $A = (V \wedge S) \vee (\neg C \wedge S_{\text{fail_safe}})$ (An Alert triggers if Vitals are high OR if Connectivity is lost).
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6. Emergency Trigger Pseudo-code

Code snippet:

```
IF HeartRate > 120 THEN
  TRY Send_Cloud_Alert()
  CATCH Connection_Error:
    Activate_Local_M2M_Mesh() // Fault-tolerance in action
END IF
```

7. References

1. *Computing*: Tanenbaum, A. S. (Modern Operating Systems).
2. *Networking*: Kurose & Ross (Computer Networking).
3. *Human/Media*: Rogers, Sharp, & Preece (Interaction Design).
4. *Convergence*: Buyya, R. (Internet of Things Principles).